

Mangrove macrobenthos: Assemblages, services, and linkages

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Abstract

Macrobenthic assemblages are relatively poorly known compared to other components of the mangrove ecosystem. Tropical mangroves support macrobenthic biodiversity resources yet to be properly documented and interpreted. Some methodological challenges, such as the generally high spatial heterogeneity and complexity of the habitat, evidently reduce sampling efficiency and accuracy, while also leaving some microhabitats under-sampled. Macrobenthic assemblage structure seems to be influenced by local environmental conditions, such as hydroperiod, organic matter availability and sediment characteristics. Brachyurans, gastropods and oligochaetes dominate in the sediment, with the former two groups also common on hard surfaces provided by tree trunks, while insects and arachnids inhabit the canopy. Traditionally, studies of mangrove macrobenthos have focused on assemblage structure or the biology of individual species, but more complex inter-specific interactions and the inter-relationship between habitat and the biota are recently being addressed. Brachyuran crabs are the best-studied macrobenthos group, but many issues about their role in mangrove ecosystem dynamics are still controversial. Despite many species of mangrove macrobenthos being referred to as ‘trophic dead ends’, most serve as important links between recalcitrant mangrove organic matter and estuarine secondary production, through feeding excursion by mobile nekton during the high tide, and macrobenthos-mediated processing and exportation of organic matter. A significant difference in the standing crop biomass of forests between the Indo-west-Pacific (IWP) and Atlantic-east-Pacific (AEP) mangroves may be related to the difference in species richness of mangrove as well as macrobenthos diversity in the two bioregions. Such differences in assemblage structure may also result in different ecosystem functioning, but the nature of the links is, however, yet to be explored. There is also a strong need for evaluating mangrove macrobenthos assemblages as a component of the connected biotic resources in the land-sea continuum, rather than as an isolated faunal group.

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1. Introduction

The macrobenthos of mangroves are relatively poorly known compared to other aspects of these productive ecosystems, such as floristics and trophic ecology. The classical account of Macnae (1968)

demonstrates, qualitatively, the diversity of the mangrove fauna, particularly the macrobenthos, in rich tropical forests in the Indo-West-Pacific, but advancement in the understanding of assemblage structure and the role of these animals in ecosystem function has since stagnated for a few decades. Mangroves have often been regarded as one of the most productive ecosystems in the world, but discussion has rarely been focussed on the contribution from the resident mangrove macrofauna.

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Table 1

The abundance and dominant taxa of macrobenthos assemblages reported from selected studies on mangroves, and factors thought to exert strong influence on abundance and structure

Location	Species richness	Numerically dominant taxa	Density (m ⁻²)	Factors shaping assemblage	Reference
Port Swettenham, Malaysia	85	Polychaetes Gastropods Crustaceans	<507 ('stream' station)	Tidal regime, sediment (grain size, pH)	Sasekumar (1974)
Sydney, Australia	17 – 29	Molluscs Crustaceans Annelids	<790	Disturbance due to boardwalk	Kelehar et al. (1998)
Kenya, Africa	ND	Oligochaetes (94%) Polychaete	23,058*	Detritus availability, predation	Schrijvers et al. (1998)
Port Hacking estuary, Australia	ND	Oligochaete Molluscs	37 – 45 per core*	No effect of urbanisation	Lindegarth and Hoskin (2001)
Mai Po, Hong Kong	46	Crustaceans Gastropods	998*	Salinity, season	Lui et al. (2002)
Puhinui Creek, New Zealand	ND	Oligochaetes Gastropods	17 – 80 per core*	Stand age (on species richness)	Morrissey et al. (2003)
Sydney, Australia	22 – 31	Oligochaetes Polychaetes	ND	No effect of sediment, high within-site variation	Chapman and Tolhurst (2004)
Matapouri estuary, northern New Zealand	29.4 per core	Oligochaetes Gastropods	8,000*	Sampling date, location	Alfaro (2006)

*- Average density for all taxa recorded in study; other values refer to maximum densities attained by individual taxa or stations. Numbers for species richness refer to totals recorded in the study, unless otherwise stated. ND - no data.

Interest in the non-fishery value of mangrove fauna rekindled in the mid 1980s when grapsid crabs were found to play an important role in leaf litter turnover (Robertson, 1986, 1991, see Robertson et al., 1992), significantly retaining mangrove organic production and reducing direct export. Subsequently, the role of the macrobenthos in shaping the structure and function of mangrove forests has been demonstrated through activities such as bioturbation, organic matter processing and propagule predation. This overview attempts to examine the current knowledge of the mangrove macrobenthos assemblages, the ecosystem services they provide and the linkage with communities and processes in adjoining habitats.

2. The assemblages

2.1. Species richness and abundance

The microhabitats offered by mangrove forests cover a wide range of coarse to fine-grained environments. Microhabitats that are occupied by macrobenthos in mangroves include the hard substrata offered by the trunk, aerial roots and foliage of mangrove trees, and the soft sediment, which is often heavily ramified by fine and shallow nutritive roots. Additional microhabitats are created for smaller benthos through the habitat modifying activities of other taxa, such as the burrows of large crustaceans. Some macrofaunal groups have been particularly successful as components of the mangrove macrobenthos, with members being important inhabi-

tants of all available microhabitats. For example, brachyuran crabs, particularly Grapsidae, display arboreal as well as epipelagic and burrowing habits (Vannini and Ruwa, 1994; Lee, 1998; Sivasothi, 2000).

Faunal assemblages of mangroves are significantly less studied and documented than the forests they inhabit. Methods for studying mangrove forest structure are well established (e.g. Cintron and Schaeffer-Novelli, 1984) and procedures for characterising the complexity and heterogeneity of mangroves are available (e.g. Lugo and Cintron, 1975). The macrobenthos of mangroves are, however, much more difficult to characterise and problems in quantifying macrobenthic abundance exist even for the most apparent groups (e.g. crabs - Nobbs and McGuinness, 1999; Skov et al., 2002).

The mangrove epibenthos are generally dominated by decapod crustaceans (especially brachyuran crabs) and gastropods, while polychaetes, oligochaetes and insect larvae make up the bulk of the infauna (Table 1). Other taxa such as sipunculids can also be locally abundant (Sasekumar and Chong, 1998). Composition of the assemblages may differ significantly depending on whether some taxa, such as nematodes, are included. Nematodes are the most abundant metazoans in estuarine sediments (Alongi, 1987; Gwyther, 2003; Gwyther and Fairweather, 2005) but most are considered to be meio-rather than macrofauna. Although traditionally attention has been focused on benthos of the sediment, the sessile epibionts of mangrove trees (e.g. barnacles) have been investigated in perspectives of recruitment and assemblage dynamics (Satumanatpan

et al., 1999; Ross, 2001; Satumanatpan and Keough, 2001), as have the diet and behaviour of tree-climbing crabs (Vannini et al., 1995; Erickson et al., 2003). Comparison with adjoining lower intertidal flats suggests shift in dominance towards taxa that are more dependent on tidal immersion, e.g. filter-feeding bivalves (Dittmann, 2002). Less is known about the functional organisation of the mangrove macrobenthos.

Studies on mangrove macrobenthos have generally reported assemblages with lower species richness than the adjoining habitats such as seagrass meadows and open sand/mudflats, but abundance levels are usually similar (Dittmann, 2001; Alfaro, 2006) (Table 1). Despite the lower species richness, abundance of mangrove macrobenthos is generally high. Natural drivers such as stand age (Morrisey et al., 2003), substrate elevation and salinity (Lui et al., 2002) have been identified as influencing assemblage structure and abundance, while anthropogenic influence such as chemical pollution and sedimentation are important in populated areas (Inglis and Kross, 2000; Skilleter and Warren, 2000; Ellis et al., 2004). High small-scale spatial variability is usually detected (Chapman and Tolhurst, 2004) while equivocal results were recorded for the effect of urban runoff on assemblage structure (Inglis and Kross, 2000; Lindegarth and Hoskin, 2001; Courtenay et al., 2005).

The mangrove habitat has a low degree of environmental favourableness, due to stresses arising from tidal fluctuations (hypersaline sediment), or hypoxia (production of toxins from reductive metabolism). Mangroves have remarkable adaptations to these conditions and are able to attain high levels of primary productivity. Part of the adaptation, however, is to minimise nutrient drain through organic matter loss. The nutritional quality of leaf litter, the largest component of mangrove organic matter input to estuarine food chains, is generally poor. This trophic limitation, together with the physical environmental stresses, may have limited the proliferation of species in the mangrove macrobenthos. The assemblages are therefore dominated by large populations of few species that are adapted to these limitations, e.g. possession of cellulases that can help digest structural polysaccharides (Neilson et al., 1986; Neilson and Richards, 1989) or an ability to synthesise polyunsaturated fatty acids from simpler precursors (unpubl. data).

2.2. Methodological issues

Many factors contribute to the methodological hurdles for quantifying macrobenthos abundance in

mangroves, one of which is structural complexity of the habitat. McCoy and Bell (1991) identified three elements of habitat structure: complexity (the absolute abundance of individual structural components), heterogeneity (the relative abundance of components) and scale (size or volume used to measure heterogeneity and complexity). Mangrove forests have no shortage of structures that define the benthic environment: prop roots, pneumatophores, knee joints create a complex epibenthic landscape, while the shallow but widespread cable and fibrous root systems result in a fine-grained and often impenetrable substrate to sample from. The spatial occurrence of these structures is highly irregular, thus contributing to both the complexity and heterogeneity of the habitat. These structures prevent the use of most devices commonly deployed for quantitative sampling of the benthos, such as trawls, coring devices, and pop nets; also making direct counting difficult. As a result of these barriers, indirect and inaccurate methods are often used, e.g. counting of burrows for estimating crab density (e.g. Steinke et al., 1993; Skov and Hartnoll, 2001; Skov et al., 2002), which are known to present biases (e.g. Warren, 1990; Nobbs and McGuinness, 1999). Macrobenthic assemblage data must therefore be viewed with some caution, as these problems with sampling affect the mangroves significantly more than other structurally less complex habitats, such as mudflats.

The difficulty in sampling particular microhabitats in the mangrove environment may have contributed to the apparent low species richness. Sediments of established mangrove forests are often impenetrable due to the myriad of fine shallow roots. Selective sampling of more easily penetrable areas to coring devices would underestimate species richness, as would the omission of other structural components of the habitat. Pneumatophores, tree trunks, prop roots and leaf litter have been, for example, demonstrated to support distinct and sometimes obscure (e.g. Morton, 1976; Gillikin et al., 2001; Proches et al., 2001) faunal assemblages that might develop intricate interactions with their mangrove hosts (Ellison et al., 1996; Gwyther, 2003; Gwyther and Fairweather, 2005). In fringing mudflats, burrows made by macrofauna such as crabs also support local assemblages (Dittmann, 1996). Failure to include these microhabitats will seriously underestimate species richness.

The lack of a satisfactory approach to measuring mangrove habitat complexity and heterogeneity also hampers efforts in relating macrobenthos assemblages to habitat conditions. Traditional complexity indices developed for terrestrial forests over-rely on above-

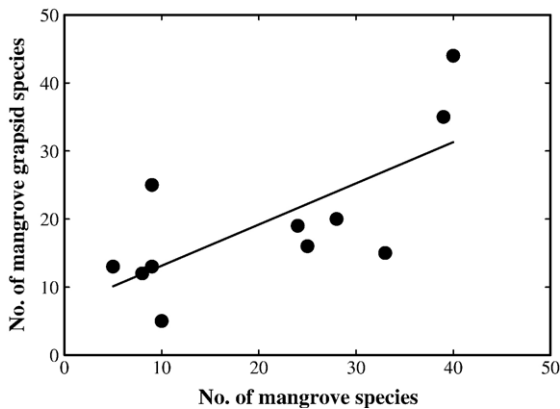


Fig. 1. Correlation between the species richness of grapsid crabs and mangroves at a regional scale. Data based on Lee (1998) with additional information from Australia (Davie, 2002), Malaysia (Ashton et al., 2003b) and Hong Kong (Lee and Leung, 1999). Regression has an r^2 value of 0.508, $p < 0.02$.

ground characteristics such as tree height and stem density (Blanco et al., 2001). These factors are relatively irrelevant to macrobenthos of mangroves, which are more responsive to forest floor level structures (e.g. prop roots, pneumatophores). Many species of mangroves produce dense root structures both above and below the sediment surface that are primarily used for assisting with respiration and anchorage. In addition to modifying the oxygen availability of surface sediment, these root structures greatly increase the complexity of the epibenthic and shallow endo-benthic environment.

Such an increase in complexity through provision of additional microhabitats should be expected to result in positive effects on biodiversity but, as mentioned earlier, no such benefits have been demonstrated for mangroves.

The spatial scale of the studies also influences the apparent pattern. Chapman (1998) and Chapman and Tolhurst (2004) attempted to relate sediment properties such as water content, colloidal carbohydrate and chlorophyll-a concentration with the abundance pattern of macrobenthic assemblages, but failed to detect correlation at any spatial scale. The latter study, however, detected significant differences in the spatial scale of variation of the sediment parameters and the benthos. It is unclear at which spatial scale macrobenthic organisms respond to broad physical habitat conditions; it is likely that relevance of physical habitat structures and conditions depends on their sizes, which would correlate positively with the size of the faunal groups concerned (Hixon and Beets, 1989, 1993). Feeding efficiency of predatory fish, for example, is dependent on habitat complexity relative to their sizes (Laegdsgaard and Johnson, 2001; Meager et al., 2005).

2.3. Species interaction

The close relationship between the macrobenthos and the mangroves can be illustrated by significant correlations between the species richness of the two. Lee (1998) reported a positive correlation between grapsid crab and mangrove species richness at a regional scale,

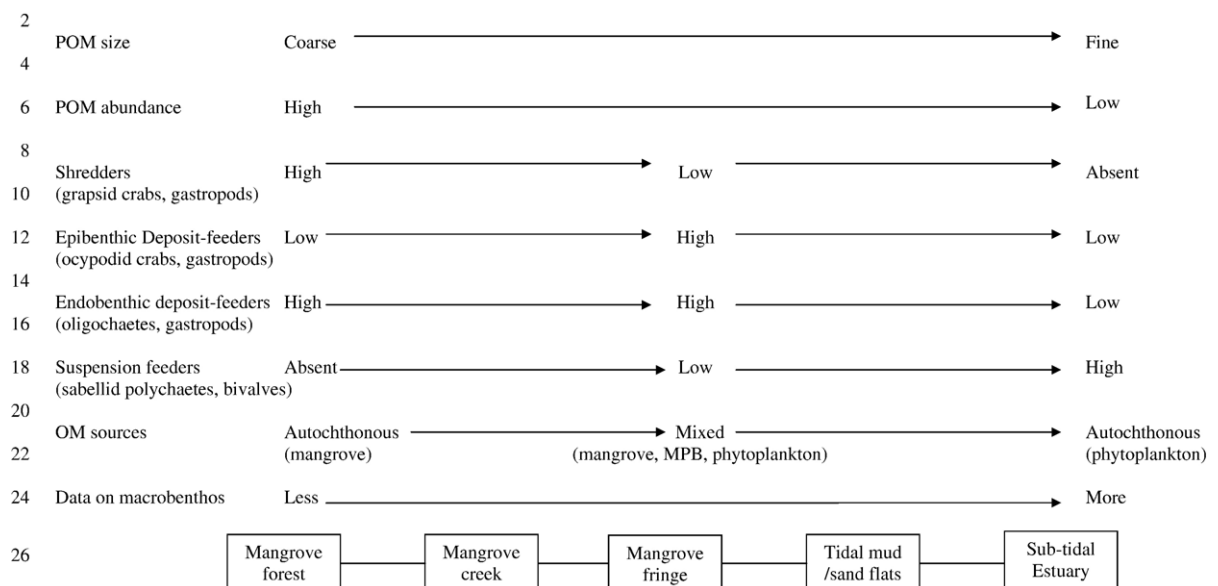


Fig. 2. A conceptual model depicting the pattern of particulate organic matter (POM) availability and utilisation in mangrove dominated estuaries, with indication of the relative abundance of the different guilds of macrobenthos involved in organic matter processing.

and the relationship seems to hold with additional recent data (Fig. 1), some of which also suggest a positive correlation at the local forest scale (e.g. Ashton et al., 2003a). The underlying explanation for such a correlation deserves more attention. It is also interesting to see at what spatial scale (e.g. local forest versus regional) the correlation is strongest, as this may shed light on the nature of the association. Correlation at the larger spatial scales is probably more a result of broad biogeographic patterns, whereas functional associations should be revealed by correlations at smaller spatial scales. A causal relationship should not be ruled out, as there is evidence of some mutual ecological dependence between mangrove and their macrobenthos, such as crabs (e.g. Smith et al., 1991; Lee and Kwok, 2002). Data on the other major macrobenthic taxa such as polychaetes and molluscs are scanty, and the lack of sufficient taxonomic resolution in most studies has further made comparisons unreliable.

In terms of functional organisation, macrobenthos associated with mangrove and adjacent environments (e.g. mud flats, subtidal estuary) are strongly influenced by the form and availability of organic matter. Large particulate organic matter (POM, e.g. leaf litter) produced by mangroves is initially processed by shredders such as grapsid crabs and gastropods in the shaded mangrove forest, where grazers of benthic microalgae are relatively rare. Shredded POM is utilised by endobenthic deposit-feeders such as oligochaetes. Tidal export of fine POM resulting from this processing then partly fuels the epibenthic consumer assemblages on the exposed banks and open sand/mud flats, dominated by deposit-feeders and grazers (e.g. ocyrodids, opisthobranchs and microgastropods), which also utilise autochthonous microphytobenthos (MPB) production (Dittmann, 2000; Dittmann and Vargas, 2001). These habitats also support high densities of deposit-feeder infauna, dominated by polychaetes and oligochaetes. Further down the tidal gradient, where MPB production is limited by water column turbidity and mangrove POM contributions are diminished, the assemblage will be dominated by filter-feeders, e.g. sabellid polychaetes and bivalves (Fig. 2).

The hostile mangrove sediment environment (low oxygen penetration, high toxins, e.g. sulfide) often limits the abundance of large infauna that do not maintain burrows to assist aeration. Some groups have, however, developed specific trophic adaptations that enable them to tap from the normally unavailable nutrient sources. Bivalves of the family Lucinidae (e.g. *Austriella*) often attain high local densities in anoxic and sulfidic mangrove sediments, their nutrition probably

assisted by their chemoautotrophic symbionts (bacteria). Their link to the mangrove food chain is, however, obscure.

Species interaction among the mangrove macrobenthos and its role in shaping faunal community structure has received little attention to date. Biotic influences such as competition and mutualism generally are more important in favourable, stable and productive habitats. Although mangroves are highly productive, environments in which they flourish cannot be described as favourable (e.g. salinity stress, high concentration of interstitial biotoxins) or stable (e.g. fluctuation in riverine input). Biotic processes are thus unlikely to exert overriding influence on community structure compared to physical factors. Activities of the macrobenthos have been shown, however, to directly or indirectly influence the other faunal components of mangroves. Numerically and functionally dominant species such as burrowing crabs have been reported to exert large influence on the distribution and abundance of other species. In most cases such influences are probably mediated through alteration of organic matter availability both spatially and in terms of quality (e.g. Lee, 1997; Botto and Iribarne, 1999), but direct evidence demonstrating such links is scanty. Such interactions could lead to secondary effects on processes such as foraging success of higher predators but the effect seems to depend on the nature of the predator-prey pair (e.g. Palomo et al., 2003; Martinetto et al., 2005). Schrijvers et al. (1995, 1998) found significant positive effects of exclusion of epibenthos on the infaunal meio- and macrobenthos, probably due to predation. Examples of facilitation have not been directly reported, but the apparent dependence on macrobenthos burrows by other invertebrates (e.g. Dittmann, 1996; Gillikin et al., 2001) suggests the existence of positive interactions. Very little is known about species interaction among the mangrove macrobenthos and much is yet to be learnt from processes shaping community structure and function in this spatially complex and productive environment. In some groups such as grapsid crabs, cannibalism can be an important process regulating species density and abundance (e.g. Kneib et al., 1999).

3. Role of macrobenthos in mangrove ecosystem services

Macrobenthos have been demonstrated to regulate the structure and function (e.g. productivity) of mangrove forests. Grapsid crabs have been proposed to influence forest structure through selective propagule predation (Smith, 1987; Osborne and Smith, 1990;

Smith et al., 1991; Frusher et al., 1994; Mcivor and Smith, 1995), with an inverse relationship between predation pressure and mangrove species dominance (the 'Predation-dominance hypothesis') but recent studies dispute the general significance of this relationship (McKee, 1995; McGuinness, 1997; Sousa and Mitchell, 1999). Manipulative experiments in northeastern Australia have demonstrated that crab predation plays a more important role in maintaining forest structure than micro-habitat condition or differences in growth between species (Clarke and Kerrigan, 2000, 2002; Clarke, 2004). At the tree level, wood borers and epibionts mostly exert negative impacts on tree growth and productivity (Perry, 1988; Brooks and Bell, 2002; Svavarsson et al., 2002). Crab activities have also been suggested to regulate forest productivity through physical modification of the habitat (e.g. increased aeration due to burrow construction) or biological influence on nutrient turnover (e.g. Smith et al., 1991).

3.1. Organic matter dynamics and biogeochemistry

By far the most studied contribution of macrobenthos to mangrove ecosystem function is their role in organic matter dynamics. Molluscan and, particularly, decapod macrobenthos have been reported to process large quantities of mangrove leaf litter (Robertson, 1986; Slim et al., 1997; Lee, 1998; Fratini et al., 2004; Proffitt and Devlin, 2005), profoundly changing the physical and chemical nature of this organic matter. Leaf litter is the most important component of mangrove primary production that is readily available to consumers, but nutritional accessibility is often limited by poor quality. Mangrove leaf litter is a rich source of carbon, but nitrogen availability is poor (C/N ratio often >100) and feeding deterrents such as polyphenolics and tannins are abundant (Neilson and Richards, 1989), making it an unattractive organic matter source. One adaptation shown by mangrove macrobenthic detritivores is that these animals seem to have developed an ability to profitably consume this nutritively poor food source, and in the process turning the material into a form that is more conducive to further consumption by themselves and other animals. Grapsid crabs, for example, shred large amounts of fresh litter in the Indo-Pacific mangroves, transforming the material partly into crab biomass, and produce faecal material with a much increased surface area to volume ratio that will facilitate microbial colonisation (Lee, 1997; Werry and Lee, 2005). In this respect, these macrobenthic consumers of fresh mangrove leaf litter perform a role intermediate between the shredders in low order streams and the

deposit-feeders in subtidal marine environments in terms of their contribution to organic matter flow and food generation for other consumers in the adjoined habitats, e.g. estuaries (Fig. 2).

Stable isotope and other tracer studies indicate that the crabs, as with most other mangrove animals, apparently do not depend heavily on the mangrove leaf organic matter for their nutritional needs (Lee, 2005; Hall et al., 2006). Many hypotheses have been proposed to explain this paradox (e.g. Skov and Hartnoll, 2002; Thongtham and Kristensen, 2005) but the most effective supplemental source is through occasional predation of other invertebrates and carrion. This model suffers, however, from the fact that this pathway will not work with molluscan detritivores such as *Terebralia palustris*, which also demonstrate significant dependence on mangrove leaf litter (Slim et al., 1997; Fratini et al., 2004).

Underground processes involving mangrove macrobenthos are virtually unknown. In mudflats, the burrows of macrobenthos such as crabs house distinct associated meio- and macrofauna (Dittmann, 1996), probably supported by local organic matter sources, e.g. cached leaves and faecal material of crabs. Irrigation and passive flushing by tidal flow, and pore water circulation may help shape the micro-environment of the burrow chambers, leading to different nutrient and organic matter dynamics. Again, this is a poorly known area of mangrove macrobenthos ecology.

Coastal wetlands are generally low-energy, depositional environments promoting sedimentation of fine particles that create a substrate with poor aeration. Mangroves demonstrate numerous adaptations for countering the hypoxia associated with the protected sediment, but the role of the macrobenthos is highly significant. Burrowing macrofauna greatly increase the surface area of the sediment-air/water interface, thus altering fundamental biogeochemical processes that are sensitive to redox conditions (Fenchel, 1996; Williamson et al., 1999; Rosenberg and Ringdahl, 2005). This change may also be mediated through vertical redistribution of sediment during burrow maintenance and construction (e.g. Warren and Underwood, 1986; Bortolus and Iribarne, 1999; Botto and Iribarne, 2000). Burrows built by large infauna also greatly modify tidal and porewater flow, contributing to materials exchange (Ridd, 1996; Stieglitz et al., 2000). The latter study demonstrated that multiple openings exist for a single burrow of the grapsid crab *Sesarma messa*, and the close spacing (<20 cm) of burrow chambers results in efficient removal of salt from sediments around mangrove roots through tidal flushing. It can be expected that exchange of other bioactive materials such as

Table 2

Predictions on aspects of ecosystem function in mangroves with contrasting levels of keystone species abundance (using the grapsid crabs as an example), such as between the Indo-west-Pacific (IWP) and Atlantic-east-Pacific (AEP) mangroves

Ecosystem attributes	Low crab richness and abundance (AEP)	High crab richness and abundance (IWP)
Forest structure	Determined mainly by propagule supply and factors such as light gap occurrence - more stochastic in nature	Determined mainly by survival upon crab predation - deterministic interaction
Organic matter retention	High OM retention especially in high-intertidal forests; export of OM mainly as coarse POM	High crab consumption especially in high-intertidal forests, significant OM retention; export of OM also as shredded micro-POM
Organic matter turnover and mineralisation rates	Slow turnover (months), mainly mediated by microbial decomposition; slower mineralisation	Fast turnover (days to months); mediated by shredders and microbes; faster mineralisation
Secondary production in forest	Low	High
Forest productivity	Low, limited by phytotoxins and nutrient mineralisation (?)	High, facilitated by bioturbation effects of crabs

phytotoxins and dissolved organic matter between mangrove sediments and tidal water may also be facilitated by the same mechanism. Macrobenthos such as fiddler crabs may work synergistically with the aerating action of *Avicennia marina* pneumatophores in regulating microbe-mediated nutrient processes in mangrove sediments (Kristensen and Alongi, 2006).

A recent study on south Atlantic estuaries has revealed that sediments with high density of the crab *Chasmagnathes granulatus* had significantly more enriched $\delta^{15}\text{N}$ signatures (by 3 – 7‰) compared to no-crab areas (Botto et al., 2005). These authors attributed the difference to increased rate of denitrification mediated by the crabs and suggest that the source and turnover of N are thus altered beyond what is usually achievable through ‘external’ drivers, such as anthropogenic influences. Direct evidence of such an effect is yet to be reported.

3.2. Ecosystem services and macrobenthos diversity

The ecological implications of the striking difference in the richness and abundance of the crab fauna between the Indo-west-Pacific (IWP) and Atlantic-east-Pacific

(AEP) (the ‘Eastern’ and ‘Western’ mangroves, respectively, *sensu* Tomlinson, 1986) deserve more attention. Given the range of ecosystem services deliverable by grapsid crabs, it may be hypothesized that this difference may result in fundamental differences in energy flow efficiency, organic matter dynamics and productivity. In the AEP mangroves, there is evidence that some of the roles played by grapsid crabs in the IWP mangroves are performed by gastropods (e.g. Proffitt and Devlin, 2005). Mcivor and Smith (1995) documented significantly lower rates in mangrove leaf consumption in a Florida mangrove compared to northeast Australia, and attributed the difference to the much impoverished grapsid crab fauna in the former. There is evidence that grapsid crab involvement in detrital dynamics greatly enhances mangrove leaf litter turnover rates, accelerates microbial detrital processing (Lee, 1997, 2005; Werry and Lee, 2005), and forest productivity might be influenced by crab bioturbation (Smith et al., 1989).

Much research effort has been paid to the relationship between terrestrial biodiversity and ecosystem services and function (Knops et al., 1999; Bunker and Naem, 2006), but aquatic, particularly marine, ecosystems have received relatively less attention (Duarte, 2000; Naem, 2006). Recent studies on stream and seagrass ecosystems have demonstrated linkages between species mix and

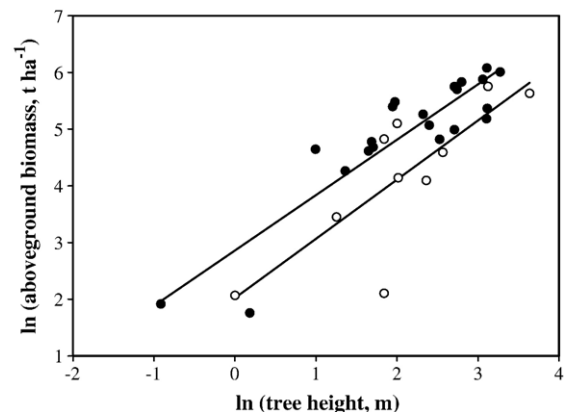


Fig. 3. The relationship between above-ground biomass and tree height in the IWP (●) and AEP (○) mangroves, based on the data from Table 5 of Fromard et al. (1998). Both variables are ln-transformed before analysis, i.e. the regression is based on the allometric relationship $\text{biomass} = a (\text{height})^b$, where a and b are constants (or $\ln(\text{biomass}) = \ln a + b \ln(\text{height})$ after ln-transformation). The regressions cover forests of the same height range and are both significant at $p < 0.001$. The slopes (b) of the regression lines for the IWP and AEP mangroves are, respectively, 0.977 and 1.043 and are equal ($p > 0.1$). The elevations ($\ln a$) of the two lines are, however, different ($t = 2.94$, $p < 0.01$), meaning that IWP mangroves have significantly higher above-ground biomass than AEP mangroves of the same tree height.

richness on ecological processes such as litter decomposition (Duarte et al., 2006; Moore and Fairweather, 2006). Both studies, however, identified stronger influence from the presence of specific species than sheer species number. The positive effect of species mixture over single species in controlling the rate of ecological processes is thought to be derived from complementarity of species roles (Hector, 1998; Hooper et al., 2005). The function of critical transitional zones such as intertidal wetlands may suffer significantly from loss of diversity (Levin et al., 2001). These authors have, however, suggested considerable levels of functional redundancy amongst the resident invertebrate fauna of estuarine and coastal wetlands. Functional redundancy is suggested to be inversely proportional to species richness in functional groups, with groups such as detritivores and deposit-feeders having high redundancy (Levin et al., 2001). This notion is, however, potentially problematic, as high diversity is often a reflection of good resource partitioning, food web complexity and thus high ecosystem stability and efficiency (Balvanera et al., 2006). The global differences in mangrove species richness have stimulated discussion on the relationship between mangrove biodiversity and ecosystem function (Field et al., 1998). There is, however, still no explicit study demonstrating this relationship to date. Predictions can similarly be made on the differences in ecosystem function resulting from the difference in species richness and abundance of important ('keystone') macrobenthos components, such as grapsid crabs (Table 2).

A preliminary comparison of the regression relationship between above-ground biomass of mangrove forests and average tree height of the IWP and AEP mangroves using the general allometric equation $\text{Biomass} = a (\text{Height})^b$ results in a significant difference in the value of a , which has a significantly higher value for the IWP mangroves (Fig. 3). As mature forests are expected to demonstrate P/B ratios ≈ 1 , this difference potentially reflects higher levels of primary productivity in the IWP mangroves. The difference could be a result of difference in climate, tree density, stand age, species architecture and anthropogenic disturbances (Clough, 1992; Clough et al., 1997; Alongi, 2002), but contrary to the conclusion of Alongi (2002), latitude was not significant in causing the difference when tree height was already in the model. This difference between the IWP and AEP mangroves is unlikely to be a result of different tree architecture but, rather, of stand characteristics, because congeners of *Rhizophora* and *Avicennia* in the IWP and AEP mangroves have comparable allometry (and thus similar mass in relation to tree height) (Clough et al., 1997). This difference may also be

a direct consequence of the diversity of the mangrove systems, demonstrating a similar relationship to other ecosystem types (Hooper et al., 2005). Since the relatively species-poor AEP mangroves still comprise genera that are the major elements in the IWP assemblage, the difference is unlikely due to difference in growth form linked to species identity. To what extent this pattern is caused by differences in ecosystem services due to different macrobenthic diversity levels in the two biogeographic regions deserves attention. A positive role of specific groups of macrobenthos, i.e. the brachyuran crabs, on mangrove productivity, has been suggested (Smith et al., 1991; Koch and Wolff, 2002). The significance of these effects is expected to vary according to factors such as local environmental gradients in salinity and human impact.

Positive species interactions, though acknowledged as common in communities and their importance highlighted in certain soft sediment environments (e.g. Reise, 2002), have been largely ignored in contemporary models of community dynamics (Bertness and Leonard, 1997). Even less is known about the importance of positive species interactions in the function (e.g. energy flow pattern) of ecosystems (Jones et al., 1997). Mangrove productivity is influenced by many factors related to drivers of geomorphic (e.g. tidal regime, hydrologic connectivity), environmental (e.g. latitude, rainfall) and ecological (e.g. macrobenthos diversity) nature, which also operate on dissimilar temporal and spatial scales (Farnsworth, 1998). It is therefore a challenging task to measure the presumably weaker effects of ecological drivers. There is no *a priori* reason why IWP and AEP mangroves should differ in how they are shaped by physical drivers; the apparent difference in biomass-tree height relationship is likely to be influenced by biotic processes related to biodiversity patterns. Issues such as functional redundancy and species functional complementarity are generally poorly documented (Hooper et al., 2005), and even less understood of macrobenthos in mangroves, and should be explored. Long-term stand-scale manipulative experiments (e.g. exclusion of key macrobenthos) conducted in different biogeographic regions with different levels of mangrove/macrofaunal diversity, and with due reference to the spatial and temporal scales of variation (Farnsworth, 1998), may contribute to assessing the effects of macrobenthos diversity on ecosystem function.

3.3. Secondary production

The contribution of mangroves to near-shore secondary production has predominantly been attributed to the

outwelling of mangrove organic matter by tidal export (Lee, 2004) while little is known about the significance of mangrove macrobenthic secondary production. Compounded by the same methodological issues limiting estimation of macrobenthos abundance, production by mangrove macrobenthos has rarely been measured and is one of the least studied areas of mangrove ecology (Alongi, 2002). Very little is known about infaunal secondary production in mangroves. Infaunal macrobenthos are probably less efficiently utilised by estuarine predators because of the limitation in access, both in terms of time and the difficulty of extraction. Schrijvers et al. (1998) suggested that since predator exclusion did not significantly benefit many infaunal macrobenthos, such as oligochaetes, they represent trophic ‘dead ends’ and contribute little to the mangrove food web. It is unclear, however, to what extent these organisms are involved in below-ground species interactions, e.g. those occurring in crab burrows. There is also the possibility that the exclusion cages did not prevent entry by intermediate predators tapping these small macrobenthos prey. The epibenthos, however, provide a significant food source to estuarine predators and may serve as an additional indirect export pathway of mangrove/marsh production to near-shore systems (Kneib, 1997; Sheaves and Molony, 2000). An evaluation of the contributions from these different components of secondary productivity may shed light on the relative significance of the short and long food chains they fuel, and the implication for estuarine nutrient dynamics.

Macintosh (1984) reported overall minimum production by *Metaplex*, *Uca* and *Sesarma* populations between $10.5 - 126 \text{ kJ m}^{-2} \text{ y}^{-1}$, with figures varying widely dependent on species. Koch and Wolff (2002) estimated epibenthic secondary productivity of a Brazilian mangrove to be $456 \text{ kJ m}^{-2} \text{ y}^{-1}$, with >95% of the biomass contributed by seven brachyuran crabs and one gastropod. Although much attention has been accorded to brachyuran crabs, given the high density (e.g. $10.5 \text{ individuals m}^{-2}$, Fratini et al., 2004) attained by gastropods in some forests, their contribution to secondary productivity should not be neglected.

The significance of the trophic link between macrobenthos and mangrove production is that some animals are adapted to digesting nutritionally poor food sources that are very low in nitrogen but high in structural carbon. Few animals have developed abilities to derive enough nutrition from such poor food sources, making the services provided by these initial processors of mangrove carbon particularly important in estuarine food chains.

One of the key issues in ascertaining the trophic link between mangrove and near-shore secondary produc-

tivity is hydrologic connectivity of the estuarine habitats, and the degree to which near-shore consumers utilise the connection for extension of their feeding habitats into the mangroves. Estuarine habitats providing good connectivity tend to support higher fish yield (Meynecke et al., 2007). In Australia, both tropical and temperate mangroves support significant presence of fish, especially juveniles (Robertson and Duke, 1990; Hindell and Jenkins, 2004; Manson et al., 2005). While feeding by these predators on mangrove macrobenthos during the high tide has not been extensively studied, similar trophic utilisation by fish has been demonstrated for Australian salt marsh habitats (Thomas and Connolly, 2001; Hollingsworth and Connolly, 2006).

4. Linkages

4.1. Mangrove-estuary linkage through animal biomass movement

Mangrove macrobenthos influence the dynamics of other adjoining ecosystems through linkages that are mediated by tidal flow, animal migrations, and direct and indirect nutrient movement and trophic relationships. Tidal flow provides the primary driving force for most material movement in mangroves, including passive movement of propagules, larval stages, litter, and vital nutrients sustaining the biota. Mangrove macrobenthos influence such material movement through their impact on organic matter retention, bioturbating activities, and also active migration. Most attention on macrobenthos-mediated material export from mangroves focussed on mangrove organic matter processing, while little is known about export in terms of animal biomass, such as eggs and larval stages. Mangrove decapod larval stages have been recorded in the plankton up to 15 km offshore (Schwamborn et al., 1999) and show salinity adaptation for this dispersal (Diele and Simith, 2006), thus contributing to the offshore food chain. Recruitment back to the adult habitat after a significant offshore planktonic larval phase is basically a stochastic process dependent on co-occurrence of conditions controlled by numerous factors, such as wind direction and current flow, as has been demonstrated for the blue crab (*Callinectes sapidus*) on the Atlantic coast of the US (Epifanio, 1995). The temporal variability of these factors is responsible for the usually large fluctuations in recruitment back to the estuarine habitats (Queiroga et al., 2006).

The potential contribution of this link to near-shore pelagic ecosystems can be estimated based on the density and fecundity of the major groups of macrobenthos

demonstrating the same life history pattern. While this information is not readily available for most groups, the brachyuran crabs provide sufficient data for an exploratory assessment. Small (<3 cm carapace width, CW, e.g. *Perisesarma* spp.) species of mangrove grapsids produce typically 10 000 – 25 000 eggs or 0.05 g AFDW per brood (Lee and Kwok, 2002; Colpo and Negreiros-Fransozo, 2003) while the fecundity of larger species (e.g. *Ucides*) may be up to 10x higher (Pinheiro et al., 2003). Mangrove grapsids can attain considerably high densities (>1 female m^{-2}) and females typically produce at least 2 – 3 broods per year (some studies in the tropics, e.g. Macintosh (1984), have reported >10 broods for some sesarmine species). The reproductive output from the brachyuran crabs can therefore be a significant indirect export pathway of mangrove organic production to supply near-shore pelagic systems. A small (20 mm CW) and relatively sparse (~ 1 individual m^{-2}) species such as *Perisesarma bidens* exports ~ 0.13 g AFDW $\text{m}^{-2} \text{y}^{-1}$ of larvae to estuarine waters and beyond. A species-rich assemblage of brachyuran crabs in tropical Australia may comprise 40 species occurring at densities averaging ~ 5 individuals m^{-2} , producing a total export of larval biomass of ~ 25 g AFDW $\text{m}^{-2} \text{y}^{-1}$. This quantity may appear small compared to particulate carbon export from mangroves (e.g. ~ 330 g C $\text{m}^{-2} \text{y}^{-1}$ for a northeastern Australian mangrove, Robertson et al., 1992) but the figures are likely to be similar in terms of N, given the large difference in the C/N ratios of animal biomass (~ 10) and senescent mangrove leaf litter (~ 100). A large proportion of the mangrove organic production consumed by these brachyuran crabs is therefore made available through this export pathway, in a form much more accessible to offshore pelagic consumers compared to senescent mangrove litter.

This link between mangrove macrobenthos and offshore communities is bi-directional. The macrobenthos of mangroves also provide crucial trophic linkage to a wide range of estuarine and coastal consumers through their occasional or regular feeding excursions into the mangroves. The degree of dependence on the macrobenthos resources, especially the infauna, by these occasional predators is still largely unknown, contributing to notions that some mangrove infauna (e.g. oligochaetes, nematodes) may be regarded as trophic ‘dead-ends’ (Schrijvers et al., 1998), i.e. the biomass is not further transferred to a higher trophic level but is only returned to the nutrient pool through natural mortality and decay. Again, this notion is probably more a consequence of the ignorance than a true representation of the nature of mangrove food webs. Juvenile fish seem to associate with mangroves because of the feeding efficiency they enjoy in

this environment (Laegdsgaard and Johnson, 2001), but there are both spatial (biogenic structures, sediment) and temporal (limited inundation time) refuges that are available to especially the infauna of mangroves. The value of the temporal refuge to the epibenthos is apparent from the behaviour that most taxa are behaviourally adapted to the tidal inundation cycle: most are active only during the low tide and would retreat to spatial refuges such as burrows or supratidal structures (e.g. tree trunk) at high tide. Individuals trying to lengthen their foraging time into the inundation period present themselves to significantly higher predation risk (Lee and Kneib, 1994).

The degree of dependence of estuarine fish and crustaceans on mangroves as feeding and nursery grounds has significant management implications, but few assessments have been made. Although juvenile fish and crustaceans undoubtedly utilise mangrove trophic resources, it is unclear if there is an obligatory dependence, i.e. mortality of the juveniles, and thus a reduction in future recruitment due to a particular cohort, would be increased when mangroves are absent. Assessing the relationship between the catch data of 31 commercially important fish species and wetland habitats in 13 independent Queensland estuaries, Meynecke et al. (2007) have found that structural connectivity of wetlands is of crucial importance to commercial catch of these species.

Linkage between the mangrove macrobenthos and coastal pelagic systems is not limited to trophic links that involve macrofauna. Dissolved material, both organic (mainly DOC) and inorganic (mainly DIN), originated from the macrobenthos in salt marsh has been demonstrated to enhance pelagic bacterioplankton production in estuarine environments (Hopkinson et al., 1998). Similar linkages are emerging from recent studies of mangroves (Dittmar and Lara, 2001; Dittmar et al., 2001, 2006), with influence of mangrove DOM extending to offshore oceanic environments.

5. Conclusive remarks

Due to difficulties in sampling and quantifying the mangrove habitat, the diversity and abundance of mangrove macrobenthos are yet to be fully documented. Some members of the mangrove macrobenthos, such as brachyuran crabs, have been demonstrated as important functional components in mangroves, particularly in terms of organic matter and nutrient processing, through their feeding and bioturbation activities. To what extent biogeographic and anthropogenic differences in macrobenthic diversity may affect mangrove ecosystem function and services presents a significant research question.

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